B.Tech (Computer Engineering), Honors 5th Semester Examination 2025-26

Data Analytics
Paper Code: CSH - 511

Max. Marks: 45 Time: 3 Hours

Instruction to the candidates:
Attempt Any Two parts from each question.
Each part of the question carries 4.5 marks.
Missing data if any may be suitable assumed.

1(a) Define the following terms: CO-1

i) Normal distribution curve BL-2
ii) Random variable
iii) Cosine similarity
iv) Mean absolute deviation

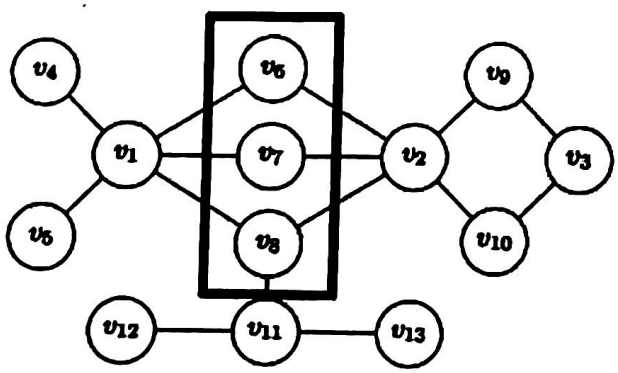
(b) What are the different types of probability distribution function? Briefly describe any two. CO-1 BL-2

(c) What is hypothesis testing? Explain Null hypothesis and alternate hypothesis. CO-1

Analyze the below data and tell whether you can conclude that smoking causes cancer or not? BL-3

Category	Diagnosed as Cancer	Without Cancer	Total
Smokers	400	300	700
Non-Smokers	300	500	800
Total	700	800	1500

2.(a) Given below is a graph. Calculate the betweenness centrality and closeness centrality of v_6, v_7 and v_8 only. CO-2 BL-4



(b) What are hubs and authority scores? Using a small example show how these values calculated.

(c) Explain the Principal component analysis used for dimensionality reduction. CL
You are given the following 3 observations. Perform PCA to find the principal components and the transformed data. BL-4

(2,0) ; (0,1); (1,3)

3(a) Given the DNA sequence database find all frequent sequences using minsup=4. Show the steps. CO-3
BL-3

S1 : AATACAAC; S2 : GTATGTGAT; S3 : ACATGGCCA; S4 : AGCGTTA

(b) i) We generally will be more interested in association rules with high confidence. However, often we will not be interested in association rules that have a confidence of 100%. Why? Then specifically explain why association rules with 99% confidence may be interesting (i.e., what might they indicate)? CO-3
BL-3

ii) Let 'C' be the set of all closed frequent itemsets and 'M' the set of all maximal frequent itemsets for some database. Prove that $M \subseteq C$.

(c) Given below is a transaction dataset. Find all Closed and Max frequent itemsets Assume MinSupport = 3. CO-3
BL-3

TID	Items
T1	{A, B, C}
T2	{A, C, D}
T3	{A, B, D}
T4	{B, C}
T5	{A, B, C, D}
T6	{A, D}
T7	{B, C, D}
T8	{A, B, D}

4.(a) How does ensemble method improve the classifier accuracy? The following sample dataset contains 8 points as (x1, x2) and the output is "Class". If you use random forest classifier how will the algorithm work. Only explain the various steps and do not give the complete solution. CO-4
BL-4

Id	x1	x2	Class
1	2	3	N
2	3	2	N
3	1	3	N
4	4	3	N
5	6	7	Y
6	7	6	Y
7	8	7	Y
8	9	6	Y

(b) What is linear and non-linear regression? Give 2 examples each on the problems where you will apply these methods. CO-4 BL-4

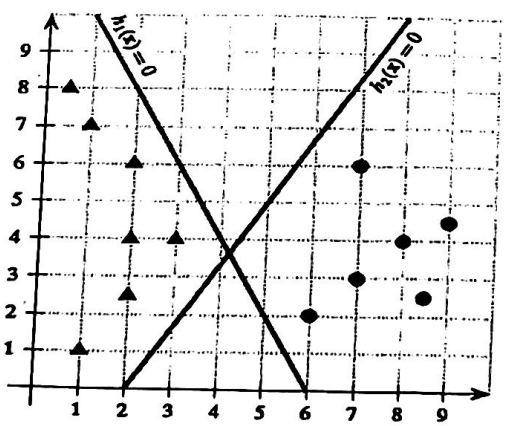
You are given the following data

Years (X)	Salary Y (In thousand)
1	30
2	40
3	50
4	55
5	70

i) Find the Regression equation
ii) Predict salary for X=6 years.

(c) Consider the dataset given below which has points from two classes c1 (triangles) and c2 (circles). Answer the question given below. CO-4 BL-4

i) Find the equation of the two hyperplanes.
ii) Show all the support vectors for h1 and h2,
iii) Which of the two hyperplanes is better at separating the two classes based on the margin computation?



5. (a) Given 2 clusters with the following points A{ (1,3), (2,6), (4,4), (5,2)} and B{(5,6), (7,3), (8,6)}.

i) Find the distance between 2 clusters using complete linkage, average linkage and centroid linkage. CO-5

ii) Which linkage will behave better in the presence of noise? Justify your answer. BL-3

(b) i) For a DBSCAN algorithm with $\varepsilon = 3$ and MinPts = 3, classify the following points into core, border, and noise: (2,3), (3,3), (4,4), (8,10), (15,30), (25,80). CO-5

BL-3

ii) If the distance between all points is equal, how will K-Means and DBSCAN clustering algorithm behave?

(c) How elbow method is used to find the optimal value of K? CO-5

BL-4

A K-means clustering result yields total SSE = 120 for K=3 and SSE = 80 for K=4.
Apply the elbow method to justify the better K value.